

Valkyrie V2 Technical Overview

Open-Source / Open-Hardware DIY High-Temperature
CoreXY Platform



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Table of Contents

INTRODUCTION	1
SPECIFICATIONS	2
System Specifications	2
Performance Specifications	4
Material Compatibility	4
1. MACHINE PLATFORM	5
FRAME	5
ENCLOSURE	6
2. MOTION SYSTEM	7
CoreXY Motion System	7
Z Motion System	8
3. BUILD PLATFORM	9
Bed System	9
Calibration System	9
4. TOOLHEAD PLATFORM	10
5. THERMAL MANAGEMENT	11
Water Cooling System	11
Chamber Heating System	11
Part Cooling System	12
6. MATERIAL MANAGEMENT	13
Filament Drybox	13
Drybox Monitoring and Control	13
Filament Feed System	14
7. ELECTRONICS & CONTROL SYSTEM	15
Main Control	15
Motor Control	15
Power System	15
Mains Input and Protection	16
Chamber Electrical Interface	16
Electronics and Component Cooling	16
Chamber Lighting	16
External Electronics Architecture	16
8. FIRMWARE & AUTOMATION	17
Firmware and User Interface	17
Motion Control	17
Thermal Management	17
Calibration	17
Safety	17
Automation Features	18
Credits and Licensing	18
Project Lead	18
Contributors	18
Licensing	18

INTRODUCTION

The Valkyrie V2 is a next-generation, open-source and open-hardware DIY high-temperature CoreXY FFF platform engineered for reliable processing of advanced engineering polymers and fibre-reinforced composites. Every major subsystem has been redesigned to increase structural rigidity, improve motion precision, enhance thermal capability, and simplify servicing while maintaining an open and modular architecture.

This document provides a technical overview of the major design changes and new features introduced in the Valkyrie V2 platform.

Safety Notice

This document is a technical overview, not a wiring or construction guide. Mains-voltage components, wiring, connectors, protective earth, fusing, insulation, and clearances must be selected and installed for the configured input voltage by a qualified person in accordance with applicable regulations and manufacturer ratings.

SPECIFICATIONS

System Specifications

Specification	Valkyrie V2
Printer Architecture	CoreXY
Build Volume	330 × 300 × 280 mm
Frame Construction	Aluminium extrusion frame
XY Motion System	CoreXY with external NEMA 17 stepper motors
XY Motors	2 × NEMA 17, 48 mm, 55 N·cm, 2.5 A, long-shaft
Z Motion System	Three independently driven belt-lift Z axes
Z Motors	3 × NEMA 17, 40 mm, 45 N·cm, 2 A
XY Positioning Resolution	10 µm/step
Z Positioning Resolution	2.5 µm/step
Linear Rails	High-temperature all-metal MGN12
Extruder	BTT V2X, 7:1 gearing
Extrusion Force	73.5 N (7.5 kgf)
Hotend	Custom water-cooled Valkyrie hotend
Maximum Nozzle Temperature	Up to 500 °C

Maximum Build Plate Temperature	Up to 150 °C
Active Chamber Temperature	Up to 110 °C verified
Chamber Heating	750 W PTC heater (RRF Chamber 0)
Chamber Lighting	3 × 350 mm, 24 V LED strips
Bed Plate	350 × 350 × 8 mm Cast Aluminium Al 5083
Bed Mounting	3-point Maxwell kinematic system with 10 mm bearing balls
Build Plate Retention	Integrated SmCo magnets
Calibration	Automatic tool probe, Z sensor, mesh bed levelling
Part Cooling	Recirculated chamber-air CPAP system
Toolhead Cooling	Closed-loop water cooling
Water Cooling	12 V P67D pump, reservoir, and 120 mm radiator
Filament Storage	Integrated heated drybox
Drybox Heating	300 W PTC heater (RRF Chamber 1)
Drybox Control	RRF Chamber 1 heater control with ESP32/DHT22 monitoring
Filament Monitoring	Integrated load cell with remaining filament weight
Filament Feeding	External filament buffer
Control Electronics	Duet 3 6HC
Firmware	RepRapFirmware
User Interface	Duet Web Control (DWC)
Standard Input Voltage	220–240 VAC
Optional Input Voltage	110–120 VAC with configuration-specific electrical components
Mains Protection	10 A fuse for the standard 220–240 VAC configuration
Machine Weight	~50 kg
<i>The Valkyrie V2 is designed around a standard 220–240 VAC electrical configuration. An optional 110–120 VAC configuration may be built using appropriately rated heaters, power supplies, switching devices, wiring, connectors, and circuit protection</i>	

Performance Specifications

Performance	Recommended	Verified
Print Speed	Up to 500 mm/s	1,000 mm/s
Acceleration	Up to 10,000 mm/s ²	10,000 mm/s ²
Volumetric Flow Rate	Material-dependent	45 mm ³ /s
Maximum Travel Speed	Up to 500 mm/s	1,000 mm/s
Chamber Heat-Up Time	—	25 min to 100 °C (from 25 °C ambient)
Chamber Temperature	Up to 110 °C	110 °C for 60 min
Drybox Temperature Range	Up to 80 °C	80 °C

Material Compatibility

Material	Status
PLA	✓
PETG	✓
ABS	✓
ASA	✓
PA/PA-CF	✓
PC/PC-CF/PC Blend CF	✓
PPS/PPS-CF/GF	✓
PSU	To be verified
PPSU	To be verified
PEI (Ultem)	To be verified
PEKK	To be verified
✓ = Verified on the current prototype. "To be verified" indicates materials that have not yet been validated The Valkyrie V2 is designed to process engineering-grade thermoplastics and fibre-reinforced composites.	

1. MACHINE PLATFORM

FRAME

- Increased frame rigidity with M6 blind joints while simplifying the assembly process
- Improved chamber sealing and cable routing by relocating the upper rear 2020 X profile
- Added a center rear 2020 Z extrusion to improve structural stiffness and support
- Redesigned the top frame structure to enable top-mounted linear rails for easier installation and servicing



Figure 1 – Valkyrie V2 frame.

ENCLOSURE

- Redesigned enclosure architecture for improved chamber sealing and reduced heat loss during high-temperature operation
- Relocated upper rear 2020 X profile to simplify panel installation, improve sealing, and provide cleaner cable routing
- Improved separation between the heated build chamber and temperature-sensitive electronics and cooling components
- Service-oriented panel arrangement providing access to motion, wiring, and thermal-management components
- External mounting of the CoreXY motors, Z motors, water-cooling radiator, pump, and part cooling blower fan to reduce exposure to elevated chamber temperatures
- Integrated routing interfaces for toolhead wiring, water-cooling hoses, CPAP ducting, and filament feeding
- Redesigned overhead-lifting door system that opens vertically instead of swinging outward, reducing the required clearance in front of the machine
- Upgraded door hinge and linkage system featuring:
 - New reinforced mounting brackets
 - 625ZZ bearings and shoulder bolts at the arm and bracket pivot points for smoother movement and reduced play
 - Improved mechanical linkage between both sides for synchronized door movement and more consistent alignment

(a) Closed door



(b) Open door



Figure 2 - Valkyrie V2 overhead door in the closed and fully raised positions. The vertically lifting mechanism provides unobstructed access to the build chamber without requiring additional clearance in front of the machine.

2. MOTION SYSTEM

CoreXY Motion System

- New external 2.5 A NEMA 17 long-shaft stepper motors with top-supported shafts for improved drive stiffness:
 - Reduced shaft deflection for improved motion consistency and repeatability
 - Improved pulley alignment and belt stability
 - Increased belt tension capability for a stiffer CoreXY drive system
 - Increased acceleration capability and torque margin
 - Improved reliability during high-temperature operation
- Reduced CoreXY drive pulleys from 20T to 16T GT2 for increased positioning resolution at 10 $\mu\text{m}/\text{step}$ and higher torque
- Upgraded all GT2 idlers to large diameter F606ZZ flanged bearings for reduced friction, improved belt guidance, less belt stress, and increased long-term reliability
- Upgraded AB motion system to all-metal high-temperature MGN12 linear rails for improved rigidity and reliability in elevated chamber temperatures
- Redesigned AB and Y motion components for improved rigidity, belt alignment, and serviceability

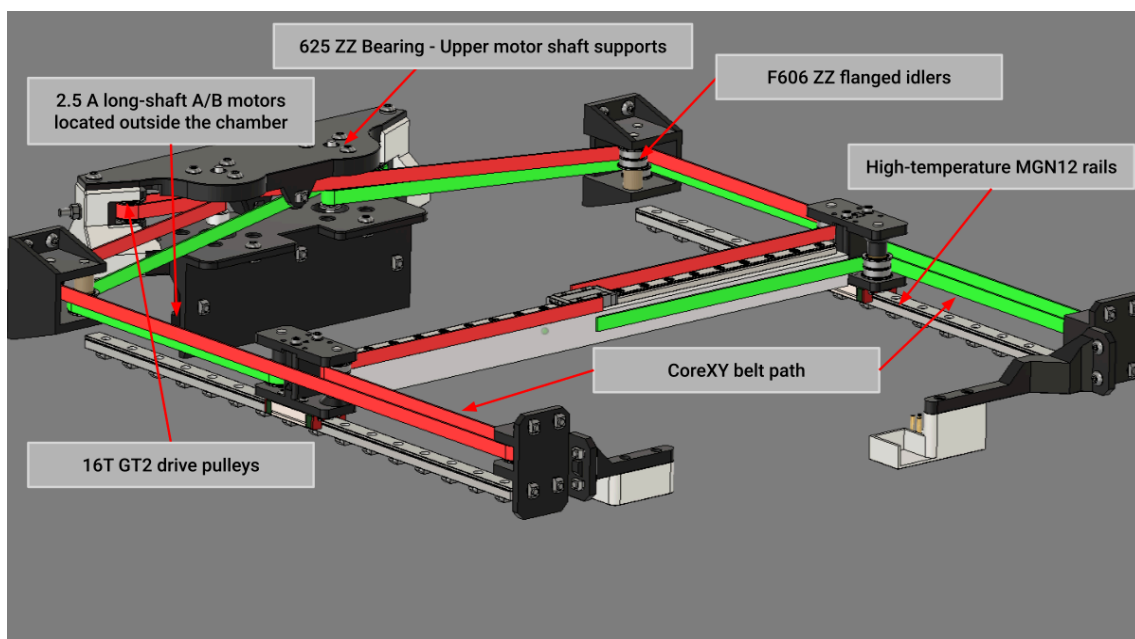


Figure 3 – Valkyrie V2 CoreXY motion system showing the external long-shaft A/B motors, upper shaft supports, 16T drive pulleys, F606ZZ idlers, and high-temperature MGN12 linear rails.

Z Motion System

- Redesigned Z carriers for increased rigidity, improved belt clamping, and ease of servicing
- New top-mounted Z idler brackets for maximum toolhead travel and optimized belt routing
- Increased Z drive reduction from 4:1 to 5:1 using 16T GT2 pulleys for higher positioning resolution at 2.5 $\mu\text{m}/\text{step}$ and increased torque
- Upgraded Z belt idlers to GT2 pulleys on 625ZZ bearing shafts for improved belt guidance and reduced friction
- High-temperature all-metal MGN12 linear rails for improved reliability in elevated chamber temperatures

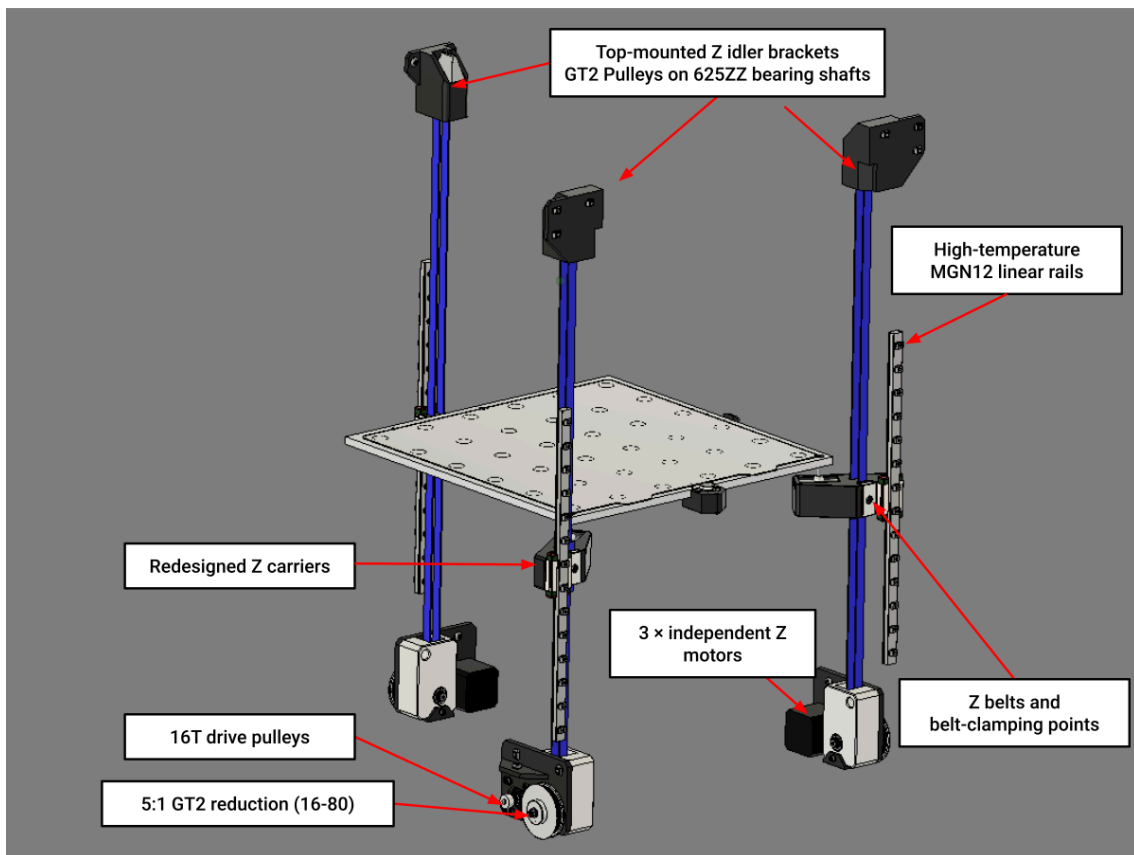


Figure 4 – Three-axis Z motion system featuring independent NEMA 17 motors, 5:1 GT2 reduction, top-mounted idlers, redesigned Z carriers, and high-temperature MGN12 linear rails.

3. BUILD PLATFORM

Bed System

- Integrated SmCo magnets for secure build plate retention and improved high-temperature stability
- Redesigned bed wiring with improved strain relief and integrated thermal fuse for increased electrical safety
- Added locating pins for repeatable build plate positioning and alignment
- 3-point independent Maxwell kinematic bed system using 10 mm bearing balls for high stability and repeatable bed positioning

Calibration System

- New Valkyrie tool probe system for automatic Z-offset measurement and nozzle height calibration
- Bed-mounted nozzle/Z-offset sensor provides a fixed reference for automatic nozzle-height measurement and Z-offset calibration
- Improved calibration repeatability for consistent first-layer performance

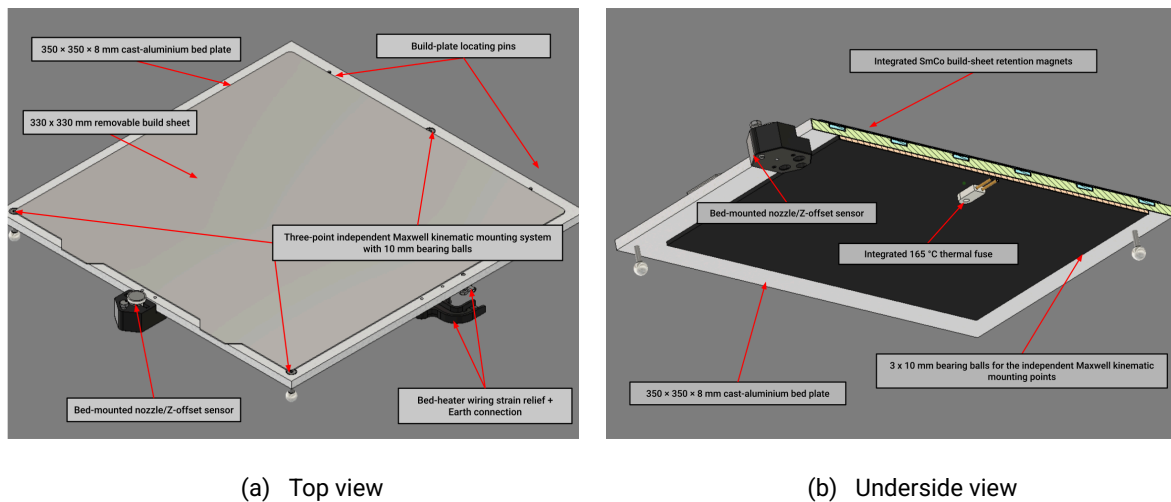


Figure 5 – Valkyrie V2 build platform: (a) 330 × 330 mm removable build sheet, cast-aluminium bed plate, SmCo magnetic retention, locating pins, and bed-mounted nozzle/Z-offset sensor; (b) three-point Maxwell kinematic mounting system, heater wiring, strain relief, and thermal fuse.

4. TOOLHEAD PLATFORM

- New water-cooled BTT V2X high-torque extruder featuring 7:1 gearing and 73.5 N (7.5 kgf) extrusion force for reliable high-force extrusion and high-flow material processing
- Custom water-cooled high-temperature hotend developed specifically for the Valkyrie V2 toolhead, featuring an integrated toolplate mounting interface for improved rigidity, alignment, and serviceability
- Redesigned integrated toolhead cable management featuring:
 - Top-mounted cable chimney routing through the toolhead cover
 - Integrated side cable channel for cleaner cable routing and improved protection
- Redesigned Z probe system with SmCo magnets for improved thermal stability and repeatability
- New magnetic Z probe docking station with SmCo magnets for secure probe retention and consistent positioning
- New nozzle brushing station for improved nozzle maintenance and print reliability

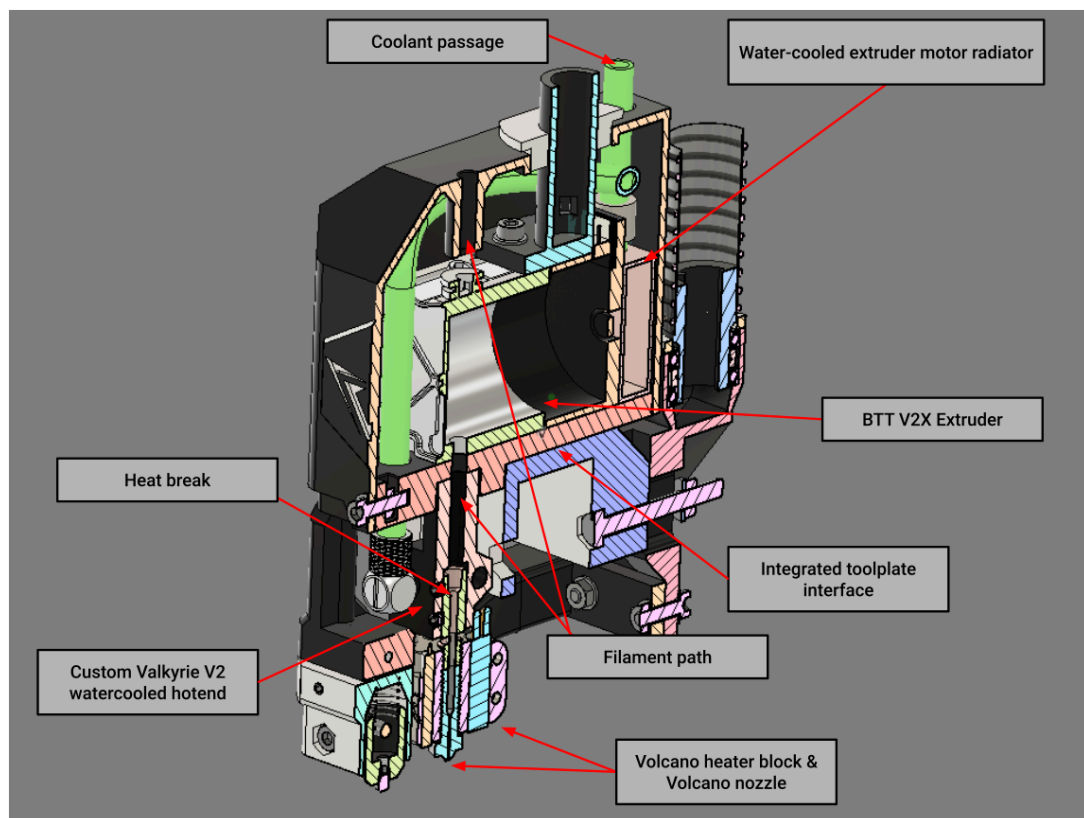


Figure 6 – Section view of the Valkyrie V2 toolhead showing the filament path, extruder drive, integrated mounting interface, and water-cooled high-temperature hotend assembly.

5. THERMAL MANAGEMENT

Water Cooling System

- Closed-loop water-cooling system using a 12 V P67D pump with water reservoir
- Dedicated cooling circuit for the extruder motor and hotend
- Integrated reservoir with fill and bleed ports for simplified coolant filling and air removal
- 120 mm radiator mounted outside the heated chamber for efficient heat rejection
- PWM-controlled radiator fan
- Cooling system isolated from the heated chamber environment

Chamber Heating System

- Dedicated 220 V, 750 W PTC chamber heater positioned centrally at the rear of the build chamber
- 220 V, 750 W heated bed provides an additional secondary heat source during chamber heating
- High-flow 120 CFM, 12 V, 92 × 92 × 38 mm circulation fan distributes heated air throughout the chamber
- Fully recirculated airflow maintains a uniform chamber temperature without introducing cooler external air
- Temperature sensor positioned at nozzle level for representative chamber-temperature measurement
- Controlled chamber operation verified up to 110 °C for processing high-temperature engineering polymers; sustained operation verified at 110 °C for 60 min
- Integrated thermal switch provides independent hardware over-temperature protection
- Firmware-configured temperature limits and automatic heater shutoff provide additional thermal safety

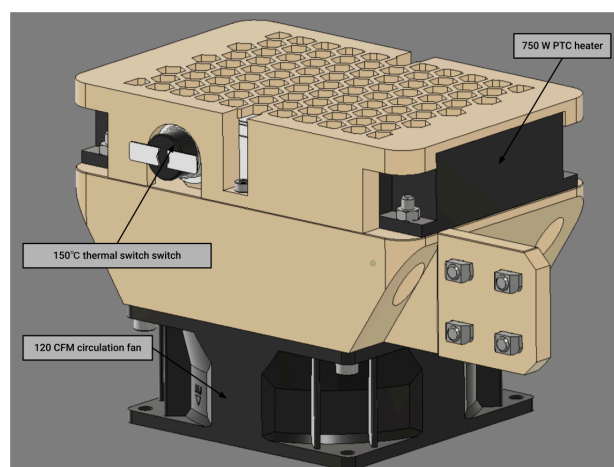


Figure 7 – Chamber-heating assembly showing the 750 W PTC heater, 120 CFM chamber-circulation fan, and independent 150 °C thermal safety switch.

Part Cooling System

- High-flow part-cooling system using an external 12 V ROBO CPAP fan
- Recirculates heated chamber air instead of introducing cooler external air, minimizing thermal shock during high-temperature printing
- Directs controlled airflow through the toolhead chimney and duct system onto the printed part
- Improves overhang and bridging performance when processing high-temperature engineering polymers
- Bearing-supported CPAP hose swivel allows free hose rotation and reduces mechanical load on the moving toolhead
- Adjustable fan control enables part-cooling airflow to be matched to the material and printing conditions

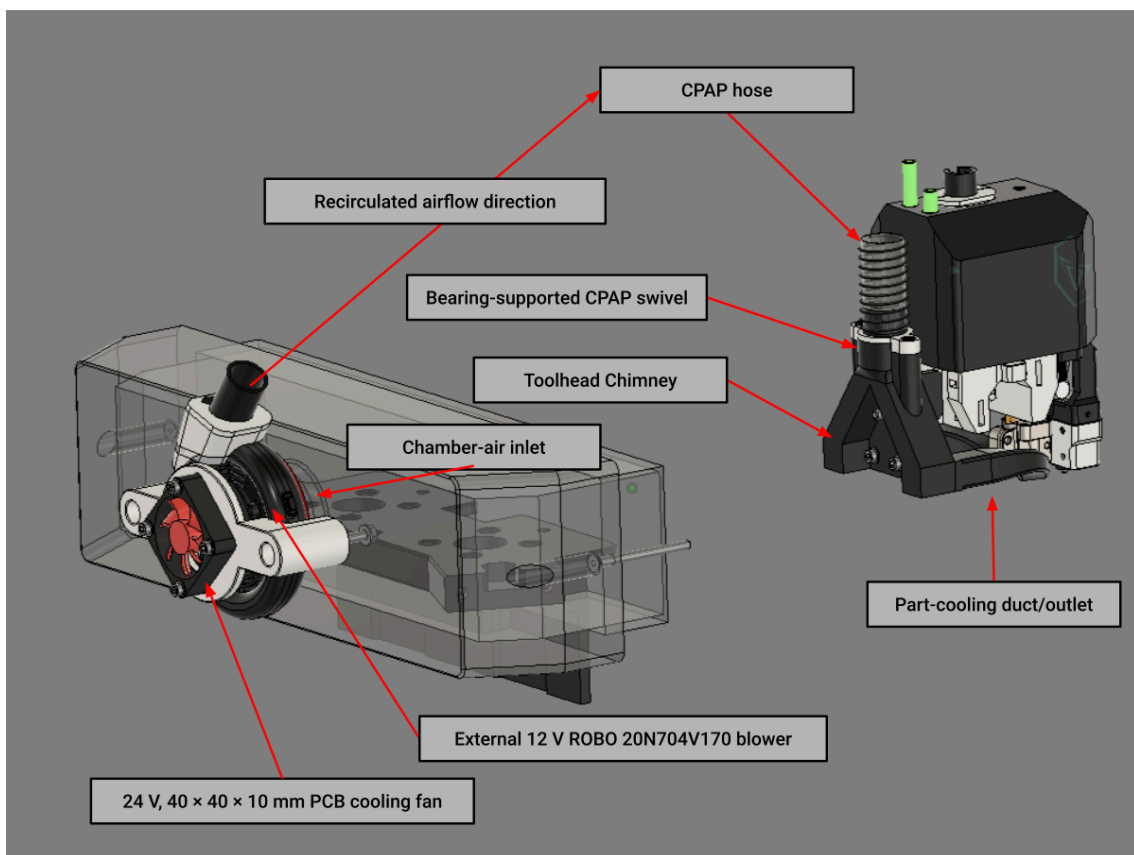


Figure 8 – Valkyrie V2 recirculated-air part-cooling system showing the chamber-air inlet, external ROBO blower, dedicated control-PCB cooling fan, CPAP hose, bearing-supported swivel, and toolhead duct directing heated chamber air onto the printed part.

6. MATERIAL MANAGEMENT

Filament Drybox

- Integrated enclosed filament drybox for moisture-sensitive engineering materials
- Dedicated 300 W PTC heater with independent temperature control
- Integrated desiccant container for passive moisture absorption and low-humidity filament storage
- Integrated humidity sensor for continuous drybox monitoring
- Integrated load cell for continuous filament-weight measurement
- Bearing-supported filament carousel using 625ZZ bearings for low-friction rotation and reliable filament feeding
- Drybox operating temperature up to 80 °C

Drybox Monitoring and Control

- RepRapFirmware directly controls the 300 W drybox PTC heater through a dedicated solid-state relay
- The drybox heater is configured in Duet Web Control as Chamber 1
- Drybox temperature setpoint, heater status, and temperature regulation are managed through DWC
- ESP32 monitoring reads the DHT22 temperature/humidity sensor and load cell
- The load cell provides continuous measurement of remaining filament weight
- The ESP32 operates as a monitoring system and does not switch or regulate the drybox PTC heater

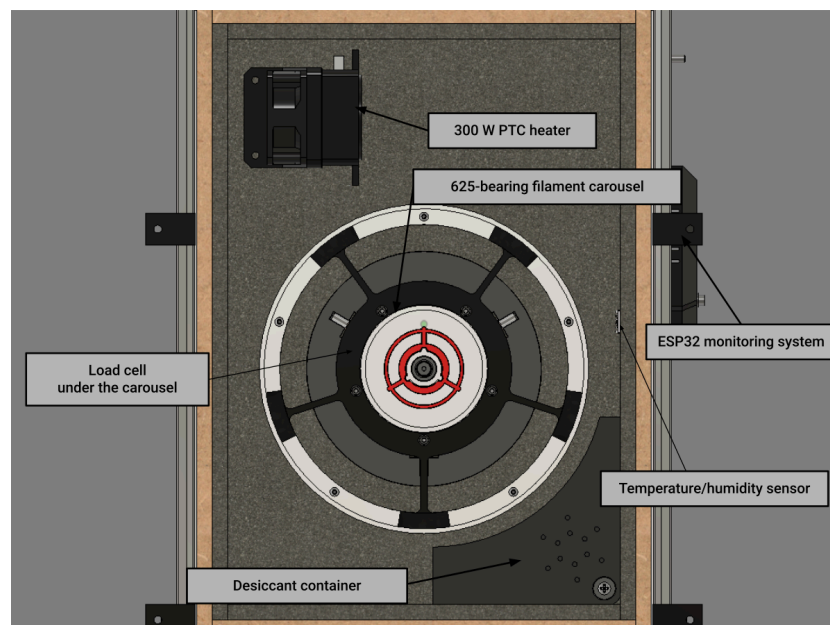


Figure 9 – Integrated filament drybox showing the heated and monitored storage system, desiccant container, load cell, and bearing-supported filament carousel

Filament Feed System

- External filament buffer positioned between the drybox and toolhead
- Isolates the extruder from spool inertia and drybox feed resistance
- Maintains consistent filament delivery during high-speed printing
- Supports automatic filament loading and unloading directly through the buffer
- Provides controlled filament storage and movement during loading, printing, and unloading operations

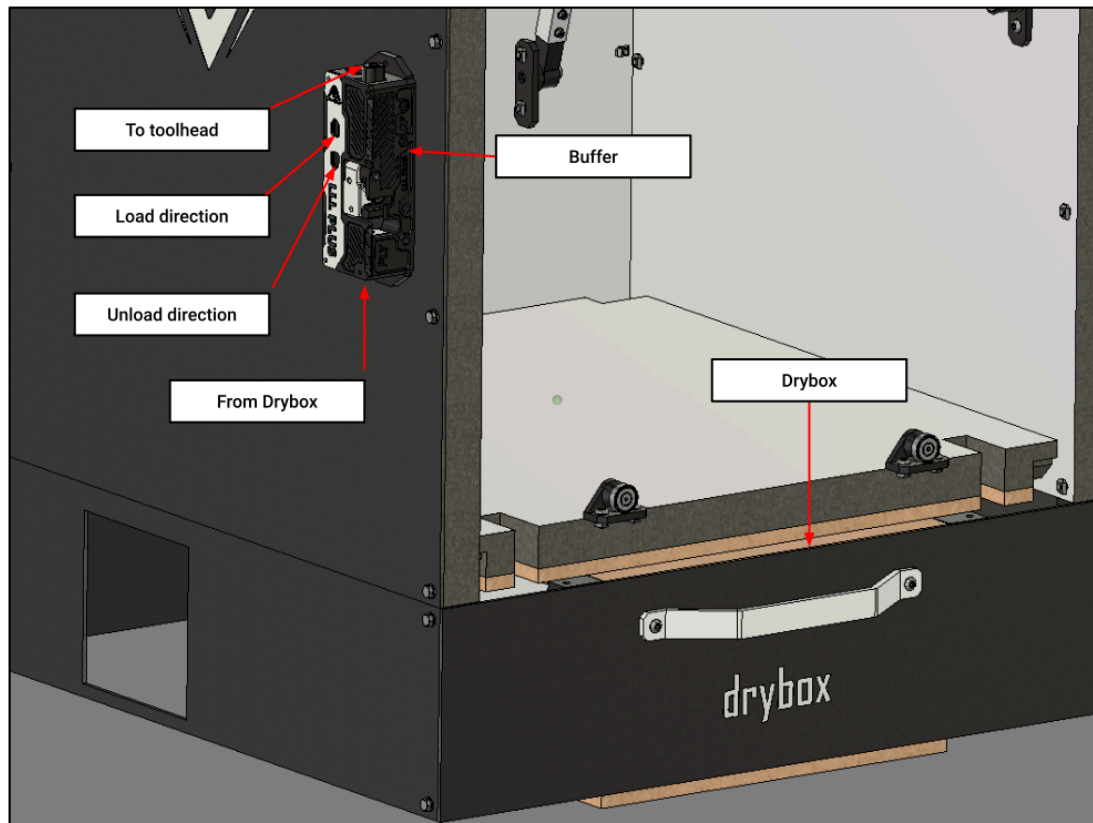


Figure 10 – External filament buffer providing automatic filament loading and unloading while isolating the toolhead extruder from spool inertia and drybox feed resistance.

7. ELECTRONICS & CONTROL SYSTEM

Main Control

- Duet 3 6HC main controller with six integrated stepper-driver channels
- No Duet expansion boards or CAN-connected toolhead electronics
- Controller mounted on the rear panel outside the heated build chamber for improved thermal reliability and service access
- All primary control and power-distribution electronics are located outside the heated chamber
- GX20 panel connectors positioned immediately outside the chamber provide the toolhead electrical interface

Motor Control

- Two independently driven CoreXY A/B motors:
 - NEMA 17
 - 48 mm body length
 - 55 N·cm holding torque
 - 2.5 A rated current
 - Long shafts with upper support
- Three independently driven Z motors:
 - NEMA 17
 - 40 mm body length
 - 45 N·cm holding torque
 - 2 A rated current
- One driver channel controls the BTT V2X extruder motor
- Independent control of the three Z motors enables automatic build-platform alignment
- All six onboard Duet 3 6HC stepper-driver channels are utilized

Power System

- Primary 24 V, 350 W power supply for the controller, motion system, chamber lighting, and other 24 V components
- Secondary 12 V power supply for pumps, fans, and auxiliary 12 V components
- TS35 DIN-rail-mounted mains and 12 V distribution for organized and serviceable wiring
- Three solid-state relays with 24 V DC control inputs switch the following 220 V AC heating loads:
 - 750 W build-plate heater
 - 750 W chamber PTC heater
 - 300 W drybox PTC heater

Mains Input and Protection

- Main IEC connector provides the external AC power connection
- Dedicated main power switch
- 15 A panel-mounted fuse holder fitted with a 10 A mains fuse

Chamber Electrical Interface

- High-temperature HDC connectors provide centralized, serviceable connections between the electronics system and heated chamber
- One 6-pin HDC connector supplies the 750 W chamber PTC heater
- One 6-pin HDC connector supplies the 4-pin 120 CFM chamber circulation fan
- One 10-pin HDC connector carries connections for:
 - Build-plate heater
 - Bed thermistor
 - Z-offset/nozzle sensor
 - Protective earth
- Centralized chamber connectors simplify installation, servicing, and component replacement

Electronics and Component Cooling

- 12 V, 120 × 120 × 25 mm fan for the water-cooling radiator
- 12 V ROBO 20N704V170 blower for recirculated-air part cooling
- 12 V, 60 × 60 × 25 mm fan for drybox-heater air circulation
- 12 V, 92 × 92 × 38 mm, 120 CFM fan for chamber-heater air circulation
- 24 V, 40 × 40 × 10 mm fan for Duet 3 6HC cooling
- 24 V, 40 × 40 × 10 mm fan for ROBO-fan control-PCB cooling

Chamber Lighting

- Three internal 24 V LED strips, each 350 mm long, provide illumination of the build chamber
- Two vertically mounted LED strips are installed on the front Z profiles and directed toward the rear of the chamber
- One LED strip is mounted at the top front of the chamber and directed toward the toolhead and build area
- Combined front and overhead illumination improves visibility of the toolhead, printed part, and internal chamber components

External Electronics Architecture

- Active electronics are positioned outside the heated build chamber wherever possible. The water-cooled toolhead extruder motor and the chamber circulation fan are the only active electromechanical components operating inside the chamber.
- External component placement reduces thermal exposure and improves reliability, accessibility, and serviceability

8. FIRMWARE & AUTOMATION

Firmware and User Interface

- RepRapFirmware (RRF) provides motion control, thermal management, calibration, safety monitoring, and printer automation
- Duet Web Control (DWC) provides the primary browser-based interface for printer operation, monitoring, file management, and configuration
- Printer configuration files and macros are stored locally on the Duet 3 6HC controller

Motion Control

- Input shaping reduces vibration and motion-related surface artifacts
- Pressure advance compensates for extrusion-system pressure changes during acceleration and deceleration
- Automated axis-homing routines establish repeatable machine coordinates
- Independent control of three Z motors enables automatic build-platform alignment
- Configurable speed, acceleration, current, and motion limits for each axis

Thermal Management

- Closed-loop temperature regulation for the hotend, build plate, and heated chamber
- RepRapFirmware independently regulates the drybox PTC heater as Chamber 1, with its temperature setpoint and operating status accessible through Duet Web Control
- Print start heats the chamber and bed first, waits for the configured minimum chamber temperature, then heats the hotend to reduce unnecessary high-temperature dwell
- Chamber, drybox, and water-pump tachometer monitoring supplements heater limits and sensor-fault handling; confirmed cooling faults trigger automatic heater shutdown
- Temperature-sensor faults automatically disable the affected heater

Calibration

- Automatic three-point Z alignment using the independently driven Z motors
- Automatic mesh bed levelling compensates for build-surface variation
- Valkyrie tool-probe system measures nozzle height and Z offset
- Automated nozzle-probing sequence provides repeatable first-layer calibration
- Nozzle-brushing routine cleans the nozzle before probing and printing

Safety

- Thermal-runaway and cooling-system monitoring for firmware-controlled heaters
- Configurable minimum- and maximum-temperature limits
- Automatic heater shutdown following sensor faults, heating errors, exceeded limits, or confirmed fan/pump cooling faults
- Independent thermal switches provide hardware-level over-temperature protection
- Build-plate heater wiring includes an integrated thermal fuse
- Mains fuse protects the printer's primary electrical input

Automation Features

- Automated machine homing and three-point bed alignment
- Automatic Z-offset measurement and mesh-bed calibration
- Automatic nozzle-brushing sequence
- Automated filament loading and unloading through the external filament buffer
- Print-start sequence heats the chamber and bed first, waits for the configured minimum chamber temperature, then heats the hotend before homing and calibration
- Configurable print-end sequence for heater shutdown, toolhead parking, and filament management

Credits and Licensing

Project Lead

Roy Berntsen

Project lead, primary mechanical designer and system integrator

Contributors

- Mark Bridgewater — Design, electronics, firmware, testing and documentation
- Chris Lombardi — ESP32 firmware

Acknowledgements

- RepRapFirmware and Duet Web Control
- Duet3D and BIQU

Licensing

Beginning with Valkyrie V2.0.0-rc.1, Project Valkyrie is released as an open-source and open-hardware project. Hardware design files are licensed under CERN-OHL-S-2.0, original Valkyrie firmware and configuration under GPL-3.0-or-later, and original documentation and project media under CC-BY-4.0. Earlier tagged releases remain under their original licence terms. Third-party materials remain subject to their respective licences.